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DEFENSE COMMISSARY AGENCY (DeCA)

Standard Operating Procedure — Supply Chain Operations Division

SOP-06: DEMAND FORECASTING

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1. Purpose

This Standard Operating Procedure (SOP) establishes and standardizes the demand forecasting methodology employed by the Defense Commissary Agency (DeCA) for all Stock Keeping Units (SKUs) carried across the DeCA enterprise. The purpose of this document is to:

- Define approved forecasting models and their application criteria for all A, B, and C tier SKUs across DeCA commissary locations and distribution centers (DCs);

- Improve inventory planning accuracy and reduce forecast error, as measured by Mean Absolute Percentage Error (MAPE), across all product categories;
- Minimize product waste, markdown losses, and spoilage attributable to over-forecasting, particularly within perishable categories including Meat, Dairy, Produce, and Frozen items;
- Reduce stockout events caused by under-forecasting, thereby supporting DeCA's mission to deliver a consistent, high-quality resale experience to active-duty military members, retirees, and their families;
- Support efficient procurement and transportation planning by providing timely, accurate forward-demand signals to buyers, vendors, and logistics partners; and
- Establish clear accountability, cycle timelines, and performance thresholds for all personnel involved in the demand planning process.

This SOP is effective as of August 1, 2026, and supersedes all prior informal demand planning guidance issued by the Supply Chain Operations Division. All demand planners, category managers, buyers, and Supply Chain Analytics personnel are required to operate in full compliance with the procedures set forth herein.

2. Scope

This SOP applies to all DeCA personnel directly engaged in demand planning, procurement, category management, logistics analysis, and vendor collaboration activities. Specifically, this document governs the following personnel and organizational units:

- **Demand Planners** — including Planner IDs PLN-03 and PLN-05, responsible for category-specific forecast generation and maintenance;
- **Category Managers** — responsible for strategic oversight of category-level demand signals and promotional event coordination;
- **Buyers** — responsible for translating approved forecasts into purchase orders and vendor replenishment actions;

- **Supply Chain Analytics Analysts** — responsible for model governance, MAPE reporting, exception analysis, and system integration;
- **Distribution Center (DC) Supervisors** — responsible for validating inbound volumes against published forecasts and flagging material discrepancies; and
- **Vendor Collaboration Managers** — responsible for coordinating Collaborative Planning, Forecasting, and Replenishment (CPFR) activities with top-tier vendors.

This SOP applies across all DeCA distribution centers and commissary locations within the continental United States (CONUS) and Outside the Continental United States (OCONUS). It covers all product categories including, but not limited to: Meat, Dairy, Produce, Frozen Foods, Dry Grocery, Beverages, Health & Beauty Care (HBC), and General Merchandise. All SKUs managed within IBM Planning Analytics (TM1) fall within the scope of this procedure.

Scope Note

This SOP does not govern vendor-managed inventory (VMI) arrangements where the vendor retains sole forecasting responsibility. VMI programs are governed separately under DeCA Directive 40-13, Annex B. However, vendor-provided forecasts for VMI SKUs must be reviewed and acknowledged by the Vendor Collaboration Manager in alignment with Section 12 (CPFR) of this procedure.

3. References

The following documents, directives, and standards are referenced herein and should be consulted in conjunction with this SOP:

Reference	Title / Description
DoD 4140.01, Vol. 1-12	DoD Supply Chain Materiel Management

Reference	Title / Description
	Policy — establishes overarching inventory and demand planning policy for all DoD logistics activities, including DeCA as a DoD Field Activity.
DeCA Directive 40-13	DeCA Supply Chain Management — governs procurement, inventory replenishment, vendor relationships, and distribution operations. Annex A covers demand planning governance; Annex B covers VMI arrangements.
DeCA Demand Planning Manual, Rev. 2026	Internal operational manual providing supplemental guidance on system-specific workflows, data entry standards, and exception handling procedures within IBM Planning Analytics (TM1).
APICS CPIM Body of Knowledge (Current Edition)	Certified in Production and Inventory Management (CPIM) standards published by APICS (Association for Supply Chain Management), which define industry-standard forecasting, inventory, and replenishment concepts referenced throughout this SOP.
IBM Planning Analytics Documentation (TM1)	Official IBM product documentation for Planning Analytics / Cognos TM1, the primary forecasting engine used by DeCA. Covers cube configuration, data feeds, and forecasting function libraries.
VICS CPFR Guidelines	Voluntary Interindustry Commerce Standards (VICS) guidelines for Collaborative Planning, Forecasting, and Replenishment, referenced in Section 12 of this SOP.

4. Definitions

The following terms are defined for the purposes of this SOP. Where a term is also defined in a reference document, the definition herein shall govern for demand forecasting operations.

Term	Definition
Forecast	A quantitative estimate of future demand for a given SKU over a defined planning

Term	Definition
	horizon. Within DeCA, the standard forecast horizon is a rolling 13-week forward window, updated on a weekly basis.
Demand Signal	Any data input or external indicator used to inform or adjust a demand forecast. Demand signals include point-of-sale (POS) data, commissary sales actuals, vendor-provided sell-through data, deployment schedules, and promotional event calendars.
MAPE (Mean Absolute Percentage Error)	The primary forecast accuracy metric used by DeCA, calculated as the average of the absolute percentage differences between forecasted and actual unit demand across all SKUs in a reporting period. Formula: $MAPE = (1/n) \times \sum Actual - Forecast / Actual \times 100$. Lower values indicate higher forecast accuracy.
Forecast Accuracy	Expressed as a percentage, Forecast Accuracy = $100\% - MAPE$. The DeCA enterprise target for overall Forecast Accuracy is $\geq 97\%$. Accuracy is calculated monthly per SKU and reported in aggregate by category and by planner.
Seasonality	A recurring, predictable pattern in demand associated with specific calendar periods, military population cycles, or cultural events. Seasonal effects are quantified using seasonal indices derived from 52 weeks of historical sales data and applied per SKU-month in the forecasting model.
Trend	A long-term directional movement (upward or downward) in demand for a SKU, distinct from cyclical or seasonal fluctuations. Trend components are captured by the Holt-Winters Triple Exponential Smoothing model (see Section 6.3).
Promotional Lift	The incremental increase in demand for a SKU attributable to a planned promotional event (e.g., price reduction, end-cap placement, advertised special). Promotional lift is expressed as a multiplicative factor applied to the baseline forecast. Example: a lift factor of 1.21 implies a 21% increase in expected demand during the promotional period.
Safety Stock	A buffer quantity of inventory held above the average expected demand to protect against demand variability and supply-side uncertainty (e.g., lead time variability). Safety stock is calculated using the

Term	Definition
	formula: $\text{Safety Stock} = Z \times \sigma_d \times \sqrt{(\text{Lead Time})}$, where Z is the service-level Z-score and σ_d is the standard deviation of daily demand.
Reorder Point (ROP)	The inventory level at which a replenishment order must be triggered to avoid a stockout during the supplier lead time. Formula: $\text{ROP} = (\text{Average Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}$.
Moving Average (Simple Moving Average, SMA)	A forecasting technique that calculates the arithmetic mean of demand over a fixed historical window (n periods). DeCA uses a 12-week rolling window for C-item SMA forecasts. Each new week, the oldest data point is dropped and the newest is added.
Exponential Smoothing (ES)	A weighted forecasting method that applies an exponentially decreasing weight to historical observations, giving more importance to recent data. DeCA uses a smoothing factor (alpha, α) of 0.3 for B-tier SKUs. Formula: $F_t = \alpha \times A_{(t-1)} + (1-\alpha) \times F_{(t-1)}$.
Holt-Winters Method	Triple Exponential Smoothing, a forecasting technique that separately models three components of demand: level (α), trend (β), and seasonality (γ). DeCA parameters: $\alpha = 0.2$, $\beta = 0.1$, $\gamma = 0.3$. Applied to A-tier, high-velocity, seasonal SKUs.
Collaborative Planning, Forecasting, and Replenishment (CPFR)	A structured, industry-standard process in which trading partners (DeCA and its vendors) jointly develop demand forecasts, share sales and inventory data, and align on replenishment plans. DeCA's CPFR program is governed in accordance with VICS guidelines and is described in Section 12 of this SOP.
Planogram	A schematic diagram or visual map that specifies the placement, facing count, and shelf allocation for each SKU within a commissary aisle or section. Planogram changes (resets, deletions, new item introductions) may materially alter demand patterns and must be communicated to demand planners at least four weeks in advance to allow for forecast adjustment.

5. Roles and Responsibilities

The following table defines the roles and primary responsibilities of all personnel involved in the demand forecasting process. All individuals listed are accountable to the Director, Logistics & Distribution, for compliance with this SOP.

Role / Position	Primary Responsibilities	Authority / Escalation
Demand Planner (PLN-03, PLN-05)	Generate and maintain rolling 13-week forecasts for assigned category portfolio by 09:00 every Monday Apply appropriate forecasting model (SMA, ES, or Holt-Winters) based on SKU tier and demand pattern Apply seasonal indices and promotional lift factors Review MAPE reports and initiate root-cause analysis for Red-flagged SKUs within 5 business days Publish approved forecasts to WMS and TMS by 12:00 Monday	Escalate unresolved accuracy issues to Supply Chain Analytics Analyst; coordinate with Category Manager for event-driven adjustments
Category Manager	Communicate promotional events, planogram resets, and new item introductions to Demand Planners ≥ 8 weeks in advance Approve promotional lift factors for A-item promotions with $\geq 20\%$ lift Provide strategic demand intelligence (e.g., competitive shifts, assortment changes) Review category-level forecast accuracy monthly	Final authority on category-level forecast overrides; escalate systemic accuracy failures to Director, Logistics & Distribution
Buyer	Translate approved 13-week forecasts into purchase orders within the procurement system Confirm vendor lead	Escalate supply constraints or lead time deviations ≥ 2 days to Demand Planner and DC Supervisor immediately

Role / Position	Primary Responsibilities	Authority / Escalation
	times and communicate changes to Demand Planner Validate safety stock adequacy at reorder point trigger Flag material supply-side risks (vendor allocation, disruptions) to Demand Planner for forecast adjustment	
Supply Chain Analytics Analyst	Administer IBM Planning Analytics (TM1) forecasting models and maintain model parameters Calculate and publish monthly MAPE and Forecast Accuracy reports by SKU, category, and planner Maintain seasonal index tables and update annually (or following significant demand pattern changes) Automate exception alert emails for Red-flagged SKUs Support CPFR data sharing via Vendor Portal	System and analytical authority for model governance; escalate model failures or data integrity issues to IT and the Director, Logistics & Distribution
DC Supervisor	Review inbound volume schedules against published 13-week forecasts Flag material discrepancies between forecasted inbound and actual receipts within 24 hours Confirm DC capacity constraints that may affect replenishment timing Provide feedback on product velocity at the DC level to inform forecast validation	Escalate inbound volume discrepancies exceeding $\pm 10\%$ of forecast to the Demand Planner and Buyer
Vendor Collaboration Manager	Coordinate quarterly CPFR meetings with top 10 vendors Facilitate sharing of DeCA forward forecasts via Vendor Portal Collect and integrate vendor-provided POS and sell-	Escalate vendor non-compliance with CPFR data sharing obligations to Category Manager and Legal/Contracting

Role / Position	Primary Responsibilities	Authority / Escalation
	through data into forecasting model Lead vendor collaboration calls for A-item promotions with projected lift $\geq 20\%$ Document agreed-upon forecast adjustments following vendor sessions	

6. Forecasting Methodology

DeCA employs a tiered forecasting methodology in which the appropriate statistical model is selected based on the SKU's velocity tier classification (A, B, or C) and its historical demand pattern (stable, trending, or seasonal). The three approved forecasting models are described in Sections 6.1 through 6.3. Model assignment criteria are summarized in the Method Selection Table (Section 6.4). Deviation from the assigned model requires written approval from the Supply Chain Analytics Analyst and documentation in IBM Planning Analytics.

6.1 Simple Moving Average (SMA) — C-Tier SKUs

The Simple Moving Average (SMA) is applied to **C-tier SKUs** exhibiting stable, low-volatility demand patterns with no discernible trend or seasonal component. C-tier items typically represent slow-moving, low-velocity SKUs that contribute a smaller proportion of total commissary revenue but require consistent stock availability.

Parameters: 12-week rolling window. Each Monday, the oldest week's demand data is removed and replaced with the most recent week's actuals, maintaining a constant 12-period average.

$$\text{SMA}_t = (D_{(t-1)} + D_{(t-2)} + \dots + D_{(t-12)}) / 12$$

Where $D_{(t-n)}$ = Actual demand in week (t-n); n = 1 to 12

Rationale: The SMA model is computationally simple and performs well for SKUs with stationary demand. Its 12-week window provides sufficient historical depth to smooth out random weekly fluctuations without overreacting to short-term anomalies. For C-tier SKUs, the cost of model complexity outweighs the marginal accuracy benefit.

Limitations: SMA is not appropriate for SKUs with identifiable trends or seasonal patterns, as it will systematically under-forecast during upward trends and over-forecast during downward trends. Planners must review C-tier SKU assignments quarterly and escalate any items exhibiting emerging trend or seasonal behavior to the Supply Chain Analytics Analyst for re-tiering.

6.2 Exponential Smoothing (ES) — B-Tier SKUs

Single Exponential Smoothing (ES) is applied to **B-tier SKUs** exhibiting moderate volatility with no strong seasonal component but with some responsiveness to recent demand shifts. B-tier items are mid-velocity SKUs representing moderate revenue contribution.

Parameters: Smoothing factor alpha (α) = 0.3. This value was selected through historical back-testing across DeCA's B-tier portfolio to minimize average MAPE while maintaining forecast stability.

$$F_t = \alpha \times A_{(t-1)} + (1 - \alpha) \times F_{(t-1)}$$

Where:

F_t = Forecast for the current period t

$A_{(t-1)}$ = Actual demand in period t-1

$F_{(t-1)}$ = Forecast for period t-1

α = 0.3 (smoothing factor)

Rationale: An alpha of 0.3 places 30% weight on the most recent actual and 70% on the prior smoothed forecast. This strikes a balance between responsiveness to real demand changes and insensitivity to random noise. Higher alpha values

(approaching 1.0) would make the model more reactive but prone to over-correction; lower values would make it sluggish.

Initialization: For newly introduced B-tier SKUs with fewer than 4 weeks of actuals, the initial forecast (F_1) is set equal to the category-level average demand per week for the corresponding store cluster, as maintained in IBM Planning Analytics.

6.3 Holt-Winters Triple Exponential Smoothing — A-Tier SKUs

The Holt-Winters Triple Exponential Smoothing method is applied to **A-tier SKUs** — the highest-velocity, highest-revenue items in the DeCA assortment, which frequently exhibit both trend and seasonal demand components. A-tier items are typically commissary staples such as boneless chicken breast (DCA-10001), ground beef (DCA-10011), fresh milk, bottled water, and similar high-frequency purchase categories.

Parameters: $\alpha = 0.2$ (level smoothing), $\beta = 0.1$ (trend smoothing), $\gamma = 0.3$ (seasonal smoothing).

$$\text{Level: } L_t = \alpha \times (Y_t / S_{(t-m)}) + (1 - \alpha) \times (L_{(t-1)} + T_{(t-1)})$$

$$\text{Trend: } T_t = \beta \times (L_t - L_{(t-1)}) + (1 - \beta) \times T_{(t-1)}$$

$$\text{Seasonal: } S_t = \gamma \times (Y_t / L_t) + (1 - \gamma) \times S_{(t-m)}$$

$$\text{Forecast: } F_{(t+h)} = (L_t + h \times T_t) \times S_{(t-m+h)}$$

Where:

Y_t = Actual demand at time t

m = Length of seasonal cycle (52 weeks for annual seasonality)

h = Number of periods ahead being forecast

$\alpha = 0.2, \beta = 0.1, \gamma = 0.3$

Rationale: A lower alpha (0.2) for the level component ensures that the model does not over-react to short-term demand spikes, which are common in high-velocity meat and dairy SKUs. A conservative beta (0.1) produces a slowly evolving trend estimate, appropriate for the relatively stable DeCA customer base. A moderate gamma (0.3) allows the seasonal indices to adapt to year-over-year shifts in military population cycles, including PCS seasons and deployment rotations, without abandoning established seasonal patterns entirely.

Model Initialization: Holt-Winters models require a minimum of two full seasonal cycles (104 weeks) of historical data for initialization. For SKUs with insufficient history, the Supply Chain Analytics Analyst shall apply category-level seasonal indices as proxies until sufficient data accumulates.

6.4 Method Selection Table — SKU Tier and Demand Pattern

The following table governs model assignment. Planners must verify SKU tier classification in IBM Planning Analytics prior to each quarterly model review. Tier reclassification requires approval from the Category Manager.

SKU Tier	Velocity / Revenue Contribution	Demand Pattern	Assigned Model	Key Parameters
A — High Velocity	Top 20% of SKUs by unit volume and revenue; approx. 80% of DC throughput	Seasonal and/or trending	Holt-Winters Triple Exponential Smoothing	$\alpha=0.2$, $\beta=0.1$, $\gamma=0.3$; 52-week seasonal cycle
A — High Velocity	Top 20% of SKUs by unit volume and revenue	Stable (non-seasonal)	Exponential Smoothing (ES)	$\alpha=0.3$; planner review quarterly for emerging seasonality
B — Moderate Velocity	Middle 30% of SKUs by unit volume and revenue	Stable or moderate volatility	Exponential Smoothing (ES)	$\alpha=0.3$; 12-week minimum history
B — Moderate Velocity	Middle 30% of SKUs by unit volume and revenue	Seasonal	Holt-Winters Triple Exponential Smoothing	$\alpha=0.2$, $\beta=0.1$, $\gamma=0.3$; escalate to Analytics Analyst for setup

SKU Tier	Velocity / Revenue Contribution	Demand Pattern	Assigned Model	Key Parameters
C — Low Velocity	Bottom 50% of SKUs; low unit volume and revenue contribution	Stable / Low volatility	Simple Moving Average (SMA)	12-week rolling window; no trend/seasonal adjustment
C — Low Velocity	Bottom 50% of SKUs	Highly intermittent or lumpy	SMA with manual planner override	Planner to set fixed safety quantity; escalate to Analyst if MAPE >15%

Model Override Policy

A Demand Planner may override the system-assigned model for a specific SKU if there is documented business justification (e.g., confirmed one-time demand event, product discontinuation). All overrides must be logged in the IBM Planning Analytics override register with the planner's ID, date, and reason. Overrides lasting more than 4 consecutive weeks require Category Manager approval.

7. Forecast Cycle

DeCA operates on a **monthly rolling 13-week forecast cycle**. The forecast represents the expected unit demand for each active SKU across each commissary location and distribution center for the 13 weeks immediately following the current forecast week. The rolling structure ensures that the planning horizon is always at least 13 weeks forward-looking, supporting procurement lead time requirements and transportation lane planning.

7.1 Weekly Refresh Schedule

The forecast is refreshed on a weekly basis according to the following mandatory schedule:

Time	Action	Responsible Party	System
Monday, 06:00 PDT	Prior week actuals automatically loaded from WMS to IBM Planning Analytics (TM1). Exception alerts generated for SKUs with actuals deviation $>\pm 15\%$ vs. forecast.	Supply Chain Analytics Analyst (automated)	WMS → TM1 Integration Feed
Monday, 07:00-09:00 PDT	Demand Planners PLN-03 and PLN-05 review exception alerts, apply model updates, seasonal index adjustments, and promotional lift factors. Finalize 13-week rolling forecast for assigned category portfolios.	PLN-03 (Meat, Dairy, Produce); PLN-05 (Frozen, Dry, HBC, General Merchandise)	IBM Planning Analytics (TM1); DeCA Forecast Dashboard (Power BI)
Monday, 09:00 PDT	Forecast submission deadline. All planner-reviewed and approved forecasts locked in TM1. Automatic workflow initiated to publish to downstream systems.	PLN-03, PLN-05	IBM Planning Analytics (TM1)
Monday, 09:00-12:00 PDT	Automated publishing of approved 13-week forecast to Warehouse Management System (WMS) for DC replenishment planning and to Transportation Management System (TMS) for lane and carrier capacity planning.	Supply Chain Analytics Analyst (automated)	TM1 → WMS Integration; TM1 → TMS Integration

Time	Action	Responsible Party	System
Monday, 12:00 PDT	Forecast publication complete. Buyers and DC Supervisors notified via automated email. Forecast visible in DeCA Forecast Dashboard (Power BI) for all stakeholders.	Automated system notification	Power BI Dashboard; Email Alert System

7.2 Planner Category Portfolio Assignments

Forecast responsibility is divided between the two primary Demand Planners as follows. This assignment is reviewed annually and may be adjusted by the Director, Logistics & Distribution, based on workload, SKU count changes, or organizational need.

Planner ID	Assigned Categories	Approximate SKU Count	Primary Model(s) Used
PLN-03	Meat, Dairy, Produce (Perishables)	~3,200 active SKUs	Holt-Winters (primary, A items); ES (B items); SMA (C items)
PLN-05	Frozen Foods, Dry Grocery, Beverages, HBC, General Merchandise	~5,800 active SKUs	ES (primary, B items); Holt-Winters (A seasonal items); SMA (C items)

Mandatory Compliance Deadline

Failure to submit the approved 13-week forecast by Monday 09:00 PDT will result in the prior week's forecast being carried forward automatically by TM1 as a temporary measure. The Demand Planner must notify the Supply Chain Analytics Analyst immediately if unable to meet the submission deadline. Repeated late submissions (more than two in any rolling 8-week period) must be reported to the Director, Logistics & Distribution.

8. Seasonal Index Application

Seasonal indices are multiplicative adjustment factors applied to the base forecast for each SKU-month combination to account for predictable, calendar-driven fluctuations in demand. A seasonal index greater than 1.0 indicates a period of above-average demand; an index less than 1.0 indicates a period of below-average demand.

8.1 Seasonal Index Calculation Methodology

Seasonal indices are calculated and maintained by the Supply Chain Analytics Analyst using the following methodology:

1. **Data Collection:** Collect 52 weeks of weekly sales actuals per SKU from the WMS actuals feed. Where possible, use a minimum of two years of data (104 weeks) and average the seasonal ratios across years to reduce noise.
2. **Average Weekly Demand:** Calculate the overall average weekly demand ($\bar{D}$) across all 52 weeks.
3. **Weekly Seasonal Ratio:** For each week w , compute $\text{Seasonal Ratio}_w = D_w / \bar{D}$, where D_w is the actual demand in week w .
4. **Monthly Seasonal Index:** Average the weekly seasonal ratios within each calendar month to produce a single monthly seasonal index (SI_m) for each SKU.
5. **Normalization:** Normalize indices so that the sum across 12 months equals 12.0 (i.e., the average index = 1.0), ensuring that seasonal adjustments do not introduce systemic bias into the annual forecast.
6. **Application:** The base forecast (F_{base}) for a given SKU in a given month is multiplied by the corresponding seasonal index: $\text{Adjusted Forecast} = F_{\text{base}} \times SI_m$.

8.2 Military Population Cycle Adjustments

The DeCA customer base is predominantly composed of active-duty military members, their dependents, and retirees. Demand patterns at commissary locations are materially affected by military population cycles that do not align with typical civilian seasonal demand patterns. The following military-specific adjustments are incorporated into the seasonal index framework:

- **PCS (Permanent Change of Station) Season (May-August):** Large-scale household relocations significantly affect store-level demand composition. Locations near major installations with high PCS volumes (e.g., Fort Liberty, JBLM, Fort Cavazos) experience demand disruption and require location-specific index adjustments during this period.
- **Deployment Rotations (Variable):** Large-scale unit deployments reduce on-installation population and depress commissary demand at affected locations. Redeployment (return from deployment) generates demand surges of 15–25% above baseline for perishable and household categories. Deployment/redeployment schedules are obtained from Installation Garrison Commanders on a quarterly basis and integrated into seasonal index adjustments.
- **Holiday Periods (Thanksgiving, Christmas/New Year, July 4th, Memorial Day, Labor Day):** Holiday periods consistently generate demand surges for meat, dairy, produce, and baking items. The BBQ season (Memorial Day through Labor Day weekend) is the highest-impact promotional and seasonal period for Meat category SKUs.

8.3 Seasonal Index Table — Q3 2026 (July–September)

The following seasonal indices are approved for application in Q3 2026. Indices reflect a blend of 52-week historical ratios and military population cycle adjustments validated by the Supply Chain Analytics team. These indices are loaded into IBM Planning Analytics as of July 1, 2026.

Category	July 2026 Index	August 2026 Index	September 2026 Index	Adjustment Basis
Meat (Fresh)	1.18	1.21	1.09	Summer BBQ season; Labor Day peak; PCS influx July–August
Dairy	1.04	1.06	1.02	Moderate summer uplift; back-to-school milk demand September
Produce	1.12	1.14	1.07	Grilling accompaniments; summer fruit peak; back-to-school salads
Frozen Foods	0.96	0.97	1.01	Summer substitution toward fresh; slight recovery September
Dry Grocery	1.02	1.03	1.05	Condiments/seasonings driven by BBQ season; back-to-school uplift September
Beverages	1.22	1.19	1.08	Peak summer hydration demand; outdoor event season
HBC	1.01	1.01	0.99	Relatively stable year-round with minor summer sunscreen uplift
General Merchandise	1.08	1.06	1.03	Outdoor/BBQ accessories; back-to-school supplies September

Seasonal indices are reviewed and updated annually by the Supply Chain Analytics Analyst, with interim adjustments permissible following confirmed material changes to installation population levels or promotional calendar shifts. Any mid-cycle index

change requires written notification to PLN-03 and PLN-05 at least 72 hours prior to implementation.

9. Promotional Demand Adjustment

Promotional events represent one of the most significant sources of demand variability in the DeCA environment. Failure to accurately forecast promotional lift can result in stockouts during high-visibility sales events, which directly impacts patron satisfaction and DeCA's mission effectiveness. This section establishes the mandatory process for identifying, quantifying, and incorporating promotional demand adjustments into the 13-week rolling forecast.

9.1 Promotional Identification and Flagging

All planned promotional events are published in the **DeCA Promotional Calendar**, maintained by the Category Management Division and released **at least 8 weeks in advance** of the promotion effective date. The Promotional Calendar is accessible to all Demand Planners and Buyers through the DeCA SharePoint intranet portal.

Upon publication, the Supply Chain Analytics Analyst flags all promoted SKUs in IBM Planning Analytics with a PROMO indicator tag. Automated alerts are sent to the responsible Demand Planner (PLN-03 or PLN-05, per category assignment) requiring confirmation that promotional lift has been incorporated into the affected weeks' forecast within three business days of the alert.

9.2 Promotional Lift Factor Application

The Demand Planner determines the appropriate promotional lift factor for each promoted SKU using the following hierarchy:

7. **Historical Promotional Lift:** Primary source. The Planner retrieves the actual demand uplift recorded during the same or comparable prior promotions for the SKU from the IBM Planning Analytics historical promo actuals database. The lift factor is calculated as: $\text{Lift Factor} = \text{Average Promoted Week Demand} / \text{Average Non-Promoted Baseline Demand}$.

8. **Category Benchmark Lift:** Where historical SKU-level promotional actuals are unavailable (e.g., new item, no prior promotion), the Planner applies the category-level benchmark promotional lift factor maintained by the Category Manager. Benchmark factors are updated semi-annually.
9. **Vendor-Provided Lift Estimate:** For A-tier SKUs, vendor-provided promotional lift estimates (sourced through CPFR collaboration) may be used to supplement or validate the internal calculation, but shall not be used as the sole basis for the lift factor without Analytics Analyst concurrence.

The adjusted promotional forecast is calculated as: **Promo Forecast = Base Forecast × Seasonal Index × Lift Factor**. The lift factor is applied only to the specific weeks within the active promotional period.

9.3 Vendor Collaboration for High-Lift A-Item Promotions

For any A-tier SKU with a projected promotional lift factor of **≥1.20 (i.e., ≥20% demand uplift)**, the Vendor Collaboration Manager is required to convene a formal vendor collaboration call with the respective supplier prior to the forecast being locked. The purpose of this call is to:

- Confirm vendor production and allocation capacity to support the projected incremental demand;
- Align on delivery timing to ensure promotional inventory arrives at DCs at least 5 business days before the promotion start date;
- Document the agreed-upon order quantity and any vendor-confirmed constraints; and
- Update the Demand Planner (PLN-03 or PLN-05, as applicable) on any adjustments to the lift factor required based on vendor capacity confirmation.

9.4 Promotional Adjustment Examples — Summer BBQ Promotion, August 2026

The following examples illustrate the application of promotional lift factors for the Summer BBQ promotional event occurring in August 2026. This promotion is among

DeCA's highest-volume annual events and requires coordination between PLN-03 (Meat category), Category Management, and the top meat vendors.

Example 1: DCA-10001 — Boneless Chicken Breast (Assorted), Fresh

SKU ID	DCA-10001
Description	Boneless Chicken Breast, Assorted Weight, Fresh — Meat Category
Tier	A (High-Velocity)
Responsible Planner	PLN-03
Promotion	Summer BBQ Promo — August 2026 (Weeks 31–35)
Base Forecast (pre-lift)	Approx. 1,820 units/week (enterprise-wide)
August Seasonal Index	1.21 (Meat category, August 2026)
Promotional Lift Factor	1.21 — derived from 3-year historical BBQ promo average; validated via vendor collaboration call with primary supplier
Adjusted Promo Forecast	$1,820 \times 1.21 \text{ (seasonal)} \times 1.21 \text{ (lift)} \approx$ 2,663 units/week
Vendor Call Required	YES — Lift $\geq 20\%$; Vendor collaboration call completed July 18, 2026

Example 2: DCA-10011 — Ground Beef 80/20, Fresh

SKU ID	DCA-10011
Description	Ground Beef 80% Lean / 20% Fat, Fresh — Meat Category
Tier	A (High-Velocity)
Responsible Planner	PLN-03
Promotion	Summer BBQ Promo — August 2026 (Weeks 31–35)
Base Forecast (pre-lift)	Approx. 1,430 units/week (enterprise-wide)
August Seasonal Index	1.21 (Meat category, August 2026)
Promotional Lift Factor	1.23 — ground beef historically outperforms category average during BBQ events; category benchmark = 1.18; SKU-specific 2-year average = 1.23; vendor confirmed supply sufficiency

SKU ID	DCA-10011
Adjusted Promo Forecast	$1,430 \times 1.21 \text{ (seasonal)} \times 1.23 \text{ (lift)} \approx$ 2,128 units/week
Vendor Call Required	YES — Lift $\geq 20\%$; Vendor collaboration call completed July 18, 2026

9.5 Post-Promotion Reconciliation

Following the conclusion of each promotional event, the Demand Planner must complete a post-promotion reconciliation within 10 business days. This reconciliation shall document: (a) the lift factor applied vs. the actual lift realized; (b) any stockout events during the promotional period; (c) any excess inventory remaining post-promotion; and (d) recommended adjustments to the SKU's historical promotional lift database for future planning cycles. Reconciliation results are reviewed by the Category Manager and stored in IBM Planning Analytics.

10. Forecast Accuracy Monitoring

Continuous monitoring of forecast accuracy is fundamental to the integrity of DeCA's demand planning process. This section defines the accuracy measurement methodology, performance thresholds, reporting cadence, and corrective action requirements.

10.1 MAPE Calculation and Reporting Cadence

MAPE (Mean Absolute Percentage Error) is calculated monthly per SKU following the receipt of weekly actuals from the WMS actuals feed. The Supply Chain Analytics Analyst publishes the monthly MAPE report no later than the 5th business day of the following month. The report is distributed to PLN-03, PLN-05, both Category Managers, all Buyers, and the Director, Logistics & Distribution.

$$\text{MAPE (\%)} = (1/n) \times \sum [|\text{Actual}_i - \text{Forecast}_i| / \text{Actual}_i] \times 100$$

$$\text{Forecast Accuracy (\%)} = 100 - \text{MAPE (\%)}$$

Where n = number of forecast periods evaluated (weeks within the reporting month)

MAPE is calculated at the SKU level and then aggregated to category, planner, and enterprise levels using a unit-weighted average to ensure that high-volume SKUs drive the aggregate metric more than low-volume C-tier items.

10.2 Accuracy Performance Thresholds

Status	Forecast Accuracy Range	MAPE Range	Required Action
GREEN	$\geq 97.0\%$	$\leq 3.0\%$	No action required. Planner documents in monthly review.
YELLOW	93.0% - 96.9%	3.1% - 7.0%	Planner reviews model parameters and demand signals. Category Manager notified. Corrective plan documented within 10 business days.
RED	$< 93.0\%$	$> 7.0\%$	Immediate root-cause analysis required within 5 business days. Director, Logistics & Distribution, notified. Corrective action plan submitted within 10 business days.

10.3 Red-Flag Root-Cause Analysis Process

Upon receipt of the monthly MAPE report, the Demand Planner (PLN-03 or PLN-05, per assignment) must initiate a formal root-cause analysis for every Red-flagged SKU within **5 business days**. The root-cause analysis shall investigate the following potential contributing factors:

- Model misassignment (e.g., SMA applied to a SKU with emerging seasonality);

- Unincorporated demand signals (e.g., missed promo, planogram reset not communicated);
- Seasonal index misalignment (e.g., military population cycle shift not reflected);
- Supplier fulfillment failures causing artificial demand suppression;
- Data quality issues in the WMS actuals feed (e.g., missing transactions, duplicate records);
- New item cannibalization or substitution effects impacting adjacent SKUs; and
- Extraordinary demand events not foreseeable through standard modeling (e.g., weather events, public health incidents).

Root-cause findings and corrective actions are documented in the DeCA Forecast Exception Log, maintained in IBM Planning Analytics, and reviewed monthly by the Supply Chain Analytics Analyst.

10.4 Forecast Accuracy Report — July 2026

The following table presents the Forecast Accuracy and MAPE results for the primary SKUs tracked by PLN-03 and PLN-05 for the reporting month of July 2026. Data reflects actuals received from the WMS actuals feed for the four-week period of July 2026.

SKU ID	Description	Planner	Category	Forecast Units (July 2026)	Actual Units (July 2026)	Forecast Accuracy %	MAPE %	Status
DCA-10001	Boneless Chicken Breast	PLN-03	Meat	7,280	7,190	98.8%	1.2%	GREEN
DCA-10003	Pork Spare Ribs, Fresh	PLN-03	Meat	3,120	3,215	97.0%	3.0%	GREEN
DCA-10005	Whole Milk, 1 Gallon	PLN-03	Dairy	9,450	9,512	99.4%	0.6%	GREEN
DCA-	Shredded	PLN-03	Dairy	4,860	4,790	98.6%	1.4%	GREEN

SKU ID	Description	Planner	Category	Forecast Units (July 2026)	Actual Units (July 2026)	Forecast Accuracy %	MAPE %	Status
10007	ed Cheddar Cheese , 2 lb							
DCA-10009	Romaine Lettuce Hearts, 3-pack	PLN-03	Produce	2,640	2,801	94.3%	5.7%	YELLOW
DCA-10011	Ground Beef 80/20, Fresh	PLN-03	Meat	5,720	5,680	99.3%	0.7%	GREEN
DCA-10013	Frozen Broccoli Florets, 2 lb	PLN-05	Frozen	1,980	1,935	97.7%	2.3%	GREEN
DCA-10014	Bottled Water, 24-Pack (16.9 oz)	PLN-05	Beverages	11,340	12,190	93.0%	7.0%	YELLOW
DCA-10015	White Rice, Long Grain, 10 lb	PLN-05	Dry Grocery	3,360	3,388	99.2%	0.8%	GREEN
DCA-10016	Charcoal Briquets, 18 lb	PLN-05	General Merchandise	1,560	1,720	90.7%	9.3%	RED
DCA-10017	Eggs, Grade A Large, 18-count	PLN-03	Dairy	6,720	6,695	99.6%	0.4%	GREEN
DCA-10019	Strawberries, 1 lb Clamshell	PLN-03	Produce	2,880	3,040	94.7%	5.3%	YELLOW

Red-Flag Action Required — DCA-10016 Charcoal Briquets

SKU DCA-10016 (Charcoal Briquets, 18 lb, assigned to PLN-05) has been flagged RED for July 2026, with Forecast Accuracy of 90.7% and MAPE of 9.3%. Root-cause analysis is mandatory and must be completed by PLN-05 no later than August 8, 2026 (5 business days from report issuance August 1, 2026).

Preliminary assessment suggests the seasonal index for General Merchandise underestimated peak summer BBQ accessory demand. PLN-05 is directed to review the seasonal index assignment and promotional demand signal for this SKU and submit a corrective action plan to the Supply Chain Analytics Analyst.

Summary — July 2026: 12 SKUs evaluated. 8 Green (66.7%), 3 Yellow (25.0%), 1 Red (8.3%). Enterprise weighted MAPE for tracked SKUs: 3.1%. Overall performance is near threshold; the Yellow SKUs in Produce and Beverages, combined with the DCA-10016 Red flag, indicate elevated risk in seasonal adjustment accuracy for summer Q3 2026 items. Planners PLN-03 and PLN-05 must review all Yellow SKUs at the next Monday cycle and document plans for improvement.

11. Safety Stock Calculation

Safety stock provides a protective buffer of inventory to guard against the combined effects of demand variability and supply uncertainty (lead time variability). DeCA maintains safety stock for all A and B-tier SKUs. C-tier SKU safety stock is set at a fixed value determined by the Demand Planner in consultation with the Buyer, based on order minimum quantities and shelf space constraints.

11.1 Formula and Z-Score Service Level Parameters

$$\text{Safety Stock} = Z \times \sigma_d \times \sqrt{(\text{Lead Time})}$$

$$\text{Reorder Point (ROP)} = (\text{Average Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}$$

Where:

Z = Service level Z-score (see table below)

σ_d = Standard deviation of daily demand (units/day) — derived from 13-week rolling actuals

Lead Time = Supplier lead time in days (from PO placement to DC receipt)

$\sqrt{\text{Lead Time}}$ = Square root of lead time, converting daily demand σ to lead time period σ

11.2 Service Level and Z-Score Reference Table

Service Level Target	Z-Score	Application Within DeCA
95.0%	1.645	C-tier SKUs and low-margin dry grocery items where stockout impact is limited
98.0%	2.054	B-tier SKUs; standard service level for the majority of non-perishable and frozen SKUs
99.0%	2.326	A-tier SKUs; perishable staples (fresh meat, dairy, eggs); high patron visibility items where stockout would directly affect commissary mission

11.3 Safety Stock Calculation Examples

Example 1: DCA-10001 — Boneless Chicken Breast (A-Tier, 99% Service Level)

SKU	DCA-10001 — Boneless Chicken Breast, Fresh
Tier / Service Level	A-Tier / 99% → $Z = 2.326$
Average Daily Demand ($\bar{D}$)	260 units/day (enterprise-wide, July 2026)

SKU	DCA-10001 — Boneless Chicken Breast, Fresh
Std. Dev. of Daily Demand (σ_d)	37.2 units/day (derived from 13-week rolling actuals)
Lead Time	3 days
Safety Stock Calculation	$SS = 2.326 \times 37.2 \times \sqrt{3} = 2.326 \times 37.2 \times 1.732 \approx \mathbf{150 \text{ units}}$
Reorder Point	$ROP = (260 \times 3) + 150 = 780 + 150 = \mathbf{930 \text{ units}}$
Interpretation	When enterprise-wide on-hand inventory of DCA-10001 reaches 930 units, a replenishment order must be triggered to maintain 99% service level through the 3-day supply lead time.

Example 2: DCA-10005 — Whole Milk, 1 Gallon (A-Tier, 99% Service Level)

SKU	DCA-10005 — Whole Milk, 1 Gallon
Tier / Service Level	A-Tier / 99% $\rightarrow Z = 2.326$
Average Daily Demand ($\bar{D}$)	340 units/day (enterprise-wide, July 2026)
Std. Dev. of Daily Demand (σ_d)	22.3 units/day (derived from 13-week rolling actuals)
Lead Time	4 days
Safety Stock Calculation	$SS = 2.326 \times 22.3 \times \sqrt{4} = 2.326 \times 22.3 \times 2.0 \approx \mathbf{104 \text{ units}}$ (rounded to 80 units per shelf capacity constraint)
Approved Safety Stock	80 units — constrained by DC refrigerated storage bay capacity; shelf review documented in IBM Planning Analytics. Buyer alerted to capacity limitation.
Reorder Point	$ROP = (340 \times 4) + 80 = 1,360 + 80 = \mathbf{1,440 \text{ units}}$

Safety stock values are reviewed quarterly by the Demand Planner and updated in IBM Planning Analytics whenever lead times change by ± 1 day or when σ_d shifts by more than 15% from the baseline used for the prior calculation.

12. Collaborative Planning, Forecasting, and Replenishment (CPFR)

DeCA's CPFR program formalizes the collaborative demand planning relationship between DeCA and its top-tier vendor partners. CPFR is a structured, data-driven process in which both parties share forecasts, sales performance data, and inventory positions to jointly improve forecast accuracy, reduce supply chain inefficiencies, and prevent stockouts and overstock situations. The program is conducted in accordance with VICS CPFR guidelines and is overseen by the Vendor Collaboration Manager in coordination with PLN-03 and PLN-05.

12.1 CPFR Program Scope

The DeCA CPFR program currently encompasses the **top 10 vendors by annual sales volume**, which together represent approximately 62% of total DeCA commissary purchases. Vendor eligibility for CPFR participation is reviewed annually by the Vendor Collaboration Manager and the Director, Logistics & Distribution. Criteria for inclusion include:

- Annual purchase volume exceeding \$15 million;
- Portfolio containing ≥ 5 A-tier SKUs;
- Demonstrated data-sharing capability (EDI or API integration with DeCA Vendor Portal); and
- History of fill rate performance $\geq 96\%$ over the prior 12 months.

12.2 CPFR Meeting Cadence and Agenda

Formal CPFR collaborative sessions are conducted on a **quarterly basis** (Q1: January, Q2: April, Q3: July, Q4: October). Each session follows a standardized agenda:

10. **Prior Quarter Performance Review:** Joint review of forecast accuracy (MAPE), fill rates, stockout events, and excess inventory incidents for all CPFR-eligible SKUs.
11. **Forward Forecast Review:** Presentation of DeCA's 13-week rolling forecast by category, including seasonal index adjustments and upcoming

promotional events. Vendor provides feedback on capacity and allocation constraints.

12. **Consensus Forecast Development:** Reconciliation of DeCA's statistical forecast with vendor-provided demand intelligence. Where discrepancies exceed $\pm 10\%$, both parties document assumptions and agree on a consensus figure.
13. **Exception Resolution:** Review and resolution of forecast exceptions flagged in the prior quarter (Red and Yellow SKUs).
14. **Action Item Assignment:** All agreed actions are recorded with owner, due date, and success metric. Minutes distributed within 3 business days.

12.3 Data Sharing via DeCA Vendor Portal

Approved forecasts are made available to CPFR vendor partners through the **DeCA Vendor Portal**, a secure, role-based web interface. The following data is shared with vendors on a rolling basis:

- **Published by DeCA (weekly update):** 13-week forward forecast by SKU; promotional calendar flags; safety stock targets; DC-level demand allocation by SKU;
- **Required from Vendors (weekly upload):** Point-of-Sale (POS) data aggregated by commissary region; sell-through rates by SKU; current production and inventory positions; confirmed lead times and any temporary supply constraints.

Vendor-provided POS and sell-through data is ingested into IBM Planning Analytics via the TM1 Vendor Data Feed integration and is available to PLN-03 and PLN-05 as a supplemental demand signal during the Monday morning forecast review window (07:00–09:00). Planners are expected to review vendor-provided signals for their A-tier SKUs and document any forecast adjustments made as a result.

The Q3 2026 CPFR session was conducted in July 2026. Key outcomes included: (1) Agreement with primary meat vendor to increase DCA-10001 and DCA-10011 allocation by 18% during Weeks 31–35 (Summer BBQ Promo); (2) Vendor confirmation of a 2-day lead time extension for DCA-10014 (Bottled Water, 24-pack) through August 31, 2026, due to filling line maintenance — PLN-05 to adjust ROP accordingly; (3) New CPFR onboarding initiated for Vendor #11 pending final data portal integration testing.

13. Systems and Tools

The following systems support the end-to-end demand forecasting process. All planners must maintain active user accounts and current certification for each system relevant to their role. System access is provisioned through the DeCA IT Help Desk and requires supervisory approval.

System / Tool	Role in Forecasting Process	Key Functions Used	Access Level Required
IBM Planning Analytics (TM1) <i>Primary Forecasting Engine</i>	Core forecasting model execution, model parameter management, seasonal index storage, override logging, exception management	Forecast cube maintenance; SMA, ES, and Holt-Winters model execution; seasonal index application; MAPE calculation; override register; CPFR vendor data integration; automated WMS/TMS publish workflow	Planner: Read/Write to assigned category cubes. Analyst: Full administrative access. Category Manager: Read/Approve.
Warehouse Management System (WMS) <i>Actuals Feed Source</i>	Source of weekly sales actuals at SKU-location level; inbound receipt confirmation; DC inventory on-hand positions	Automated weekly actuals extract to TM1 (Monday 06:00); on-hand inventory query for ROP monitoring; receipt confirmation for safety stock validation	Planner: Read only. DC Supervisor: Read/Write (receipt confirmation). Buyer: Read only.

System / Tool	Role in Forecasting Process	Key Functions Used	Access Level Required
Transportation Management System (TMS) <i>Lane and Carrier Planning</i>	Recipient of published 13-week forecast for transportation lane capacity planning; inbound freight scheduling	Forecast volume receipt for carrier capacity booking; lane load planning; inbound delivery schedule generation for DC Supervisors	Planner: Read only (published forecast). TMS team: Full access.
DeCA Forecast Dashboard (Power BI) <i>Planner Review and KPI Monitoring</i>	Visual interface for planner review of 13-week forecasts, MAPE trending, SKU-level accuracy drill-down, and Red/Yellow/Green status monitoring	Forecast vs. actuals comparison charts; MAPE by SKU, category, and planner; Red-Flag SKU queue; seasonal index visualization; promotional lift overlay view	All demand planning, category management, and buying personnel: Read access. Analytics Analyst: Read/Write (dashboard maintenance).
DeCA Vendor Portal <i>CPFR Data Exchange</i>	Secure web portal for sharing DeCA forward forecasts with CPFR vendors and receiving vendor-provided POS, sell-through, and inventory data	Forecast publication to vendors; vendor data upload ingestion; CPFR meeting schedule and minutes repository; promotional calendar visibility to vendors	Vendor Collaboration Manager: Full access. Vendors: Restricted to own SKU data. Planners: Read access to vendor-uploaded data.
Email Alert System (Automated Exceptions) <i>Exception Notification</i>	Automated email generation for Red and Yellow SKU flags, Monday forecast deadline reminders, CPFR data upload compliance alerts, and safety stock breach notifications	Red-flag notification to planner + category manager; MAPE report distribution; system-generated forecast submission confirmation; safety stock breach alerts to buyer	Automated — no manual access required for standard operations. Analytics Analyst configures alert rules in TM1.

14. Key Performance Indicators (KPIs)

The following KPIs govern the performance of DeCA's demand forecasting function. KPIs are reported monthly in the Supply Chain Operations Performance Review, presented to the Director, Logistics & Distribution. Targets are set annually and reviewed as part of the SOP's annual review cycle.

KPI	Description	Target	Reporting Frequency	Owner
Overall Forecast Accuracy	Unit-weighted average Forecast Accuracy across all active SKUs, enterprise-wide. Calculated as 100% – MAPE (weighted).	$\geq 97.0\%$	Monthly	Supply Chain Analytics Analyst
MAPE — Meat Category	Mean Absolute Percentage Error for all Meat category SKUs (fresh and frozen). Primary responsibility: PLN-03.	$\leq 3.0\%$	Monthly	PLN-03
MAPE — Dairy Category	Mean Absolute Percentage Error for all Dairy category SKUs including milk, cheese, eggs, butter, yogurt. Primary responsibility: PLN-03.	$\leq 2.0\%$	Monthly	PLN-03
MAPE — Produce Category	Mean Absolute Percentage Error for all fresh Produce SKUs. Higher target threshold reflects inherent volatility of fresh produce demand.	$\leq 4.0\%$	Monthly	PLN-03
MAPE — Frozen Category	Mean Absolute Percentage Error for all Frozen Foods SKUs. Primary responsibility: PLN-05.	$\leq 3.0\%$	Monthly	PLN-05
MAPE — Dry Grocery Category	Mean Absolute Percentage Error for all Dry Grocery SKUs. Dry grocery	$\leq 2.0\%$	Monthly	PLN-05

KPI	Description	Target	Reporting Frequency	Owner
	exhibits the lowest demand volatility across DeCA categories. Primary responsibility: PLN-05.			
Red-Flag SKU Rate	Percentage of active SKUs receiving a Red (Forecast Accuracy <93%) status in the monthly MAPE report. Calculated as: (# Red SKUs / Total Active SKUs) × 100.	≤ 5.0%	Monthly	PLN-03, PLN-05
Safety Stock Adequacy Rate	Percentage of SKUs for which the approved safety stock quantity is maintained at or above the calculated target level at the time of each weekly cycle review. Measured via WMS on-hand query.	≥ 99.0%	Weekly (reviewed monthly)	Buyer, PLN-03, PLN-05
Stockout Events Attributable to Forecast Error	Count of commissary-level stockout events (zero on-hand, patron demand unsatisfied) where root-cause investigation confirms forecast under-estimation as the primary contributing factor. Excludes supply-side failures (vendor	0 events/month	Monthly	PLN-03, PLN-05, Buyer

KPI	Description	Target	Reporting Frequency	Owner
	non-delivery, transportation disruptions).			
Forecast Submission Compliance Rate	Percentage of weekly forecast cycles in which the approved 13-week forecast is submitted by the mandatory Monday 09:00 PDT deadline. Calculated per planner ID.	100%	Monthly	PLN-03, PLN-05
CPFR Vendor Data Compliance Rate	Percentage of CPFR-enrolled vendors who have uploaded required weekly POS and sell-through data to the Vendor Portal within the required timeframe (by Friday 17:00 of each week).	≥ 90.0%	Monthly	Vendor Collaboration Manager

15. Training Requirements

All personnel responsible for demand forecasting activities under this SOP must meet the training requirements specified below. Training completion is tracked by the DeCA Human Capital Division and reported to the Director, Logistics & Distribution, on a semi-annual basis. Failure to complete mandatory training within the specified timeframe may result in temporary suspension of forecasting access pending remediation.

Training Requirement	Applicable Roles	Frequency	Completion Deadline	Administering Authority
IBM	Demand Planners (PLN-	Initial (upon hire or role	Within 60 days of hire/role	IBM Learning; DeCA IT

Training Requirement	Applicable Roles	Frequency	Completion Deadline	Administering Authority
Planning Analytics (TM1) Demand Planning Certification	03, PLN-05), Supply Chain Analytics Analyst	change); Recertification every 2 years	assignment	Training Coordinator
APICS CPIM – Module: Demand Management and Forecasting	Demand Planners, Supply Chain Analytics Analyst, Category Managers	One-time completion; encouraged to pursue full CPIM certification	Within 12 months of hire/role assignment	APICS (ASCM); self-paced with DeCA tuition assistance
Annual Demand Forecasting Methodology Refresher	All personnel in Section 2 (Scope)	Annual	By September 30 each year	Supply Chain Analytics Analyst (delivers internally); reviewed and updated to reflect current SOP revision
New Planner 90-Day On-the-Job Training (OJT) Program	New Demand Planners	One-time (initial hire or transfer into planner role)	Completed within first 90 calendar days in role	Supervised by senior Demand Planner (PLN-03 or PLN-05, as applicable); OJT checklist signed off by Supply Chain Operations Division Chief
CPFR Process and Vendor Portal Training	Vendor Collaboration Manager, Demand Planners	Initial + annual refresher (30-minute online module)	Within 30 days of hire/role change; annual refresher by September 30	DeCA Vendor Collaboration Office; VICS CPFR online resources
DeCA Forecast Dashboard (Power BI) User Training	Demand Planners, Category Managers, Buyers, DC Supervisors	Initial; updated module issued following major dashboard redesign	Within 30 days of system access provisioning	Supply Chain Analytics Analyst (delivers internally); Microsoft Learn resources
FOUO / Information Handling Training	All personnel with access to DeCA-SC-SOP-06 and associated data systems	Annual (DoD-mandated)	By December 31 each year	DoD Cyber Awareness Challenge; tracked in ATCTS/JKO

15.1 New Planner OJT Program — 90-Day Milestones

The 90-day OJT program for newly assigned Demand Planners is structured as follows:

- **Days 1-15 (Orientation):** Review of all applicable SOPs, DeCA Directives, and the Demand Planning Manual Rev. 2026. System access provisioning (TM1, WMS, Power BI, Vendor Portal). Shadow senior planner through two full Monday forecast cycles.
- **Days 16-30 (Supervised Practice):** New planner independently generates 13-week forecast for a limited portfolio of 50 B and C-tier SKUs under senior planner supervision. All submissions reviewed and approved by senior planner before publication.
- **Days 31-60 (Progressive Independence):** New planner assumes responsibility for a defined subset of assigned category portfolio ($\leq 30\%$ of total SKU count). Senior planner reviews all Red and Yellow flagged SKUs. Planner completes IBM TM1 certification and APICS CPIM demand module enrollment.
- **Days 61-90 (Full Portfolio Transition):** New planner assumes full responsibility for assigned portfolio with monthly MAPE review by Supply Chain Analytics Analyst. OJT checklist completed and signed. Formal performance evaluation conducted by Division Chief at Day 90.

16. Revision History

Version	Date	Author / Prepared By	Approved By	Description of Changes
1.0	August 1, 2026	Supply Chain Analytics Team, DeCA HQ	Director, Logistics & Distribution	Initial release. Establishes the DeCA standardized

Version	Date	Author / Prepared By	Approved By	Description of Changes
				demand forecasting SOP, replacing all prior informal demand planning guidance. Covers SMA, ES, and Holt-Winters methodology; 13-week rolling forecast cycle; seasonal index framework including Q3 2026 indices; promotional lift process with Summer BBQ 2026 examples (DCA-10001, DCA-10011); forecast accuracy thresholds and July 2026 MAPE report; safety stock formula with DCA-10001 and DCA-10005 examples; CPFR program structure; IBM Planning Analytics and Power BI system framework; KPIs and training requirements. Applicable to Planner IDs PLN-03 and PLN-05.

Approval Signatures

By signing below, the undersigned certify that they have reviewed this Standard Operating Procedure and that it accurately reflects current DeCA policy and operational practice as of the effective date.

**Reviewed By: Supply Chain
Operations Division Chief** Supply Chain
Operations Division Defense Commissary
Agency Date: _____

**Approved By: Director, Logistics &
Distribution** Defense Commissary Agency
Fort Gregg-Adams, Virginia Date: _____

DeCA-SC-SOP-06 | Revision 1.0 | Effective: August 1, 2026 | Next Review: August 1, 2027

Defense Commissary Agency — Supply Chain Operations Division — Fort Gregg-Adams, Virginia

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